

IT Research Methodology

Research Problem

Image Source: Research Methodology

Prepared by: CHEN SOVANN

❖ Learning Outcome

- Define a research problem and explain its importance in a study
- Distinguish between a research problem and other parts of the research process
- Identify criteria for deciding whether a problem can and should be researched
- Describe how quantitative and qualitative research problem differ
- Learn the five elements of writing a “statement of the problem” section
- Identify strategies useful in writing a “statement of the problem” section

Outline

- Definition of Research Problem
- Sources of Research Problems
- Steps in Formulating a Research Problem
- Errors in Defining the Problem Statement
- Quantitative and Qualitative Research Problem
- Elements of Writing a “Statement of the Problem”
- Group Discussion

❖ Definition of Research Problem

➢ A research problem is

✓ A statement about an area of concern,

✓ A condition to be improved, a difficulty to be eliminated, or

✓ A troubling question

that exists in **scholarly literature**, in **theory**, or in **practice** that points to the need for **meaningful understanding** and **deliberate investigation**.

Source: Research Problem

❖ Definition of Research Problem

➢ Problem Statement

Characteristics of Problem Statement

- **Know Fact**: can be used to introduce the topic
- **Gap**: identify gap from the current solution
- **Appealing**: preserve researcher interests
- **Exploratory**: can be investigated through the collection and analysis of data
- **Noteworthy**: contribute for the improvement in the research subject
- **Feasible**: fit the researcher's level of research skills, available resources, and time restrictions
- **Ethical**: be ethical and not harm anybody
- **Challenging**
- **Clarity**

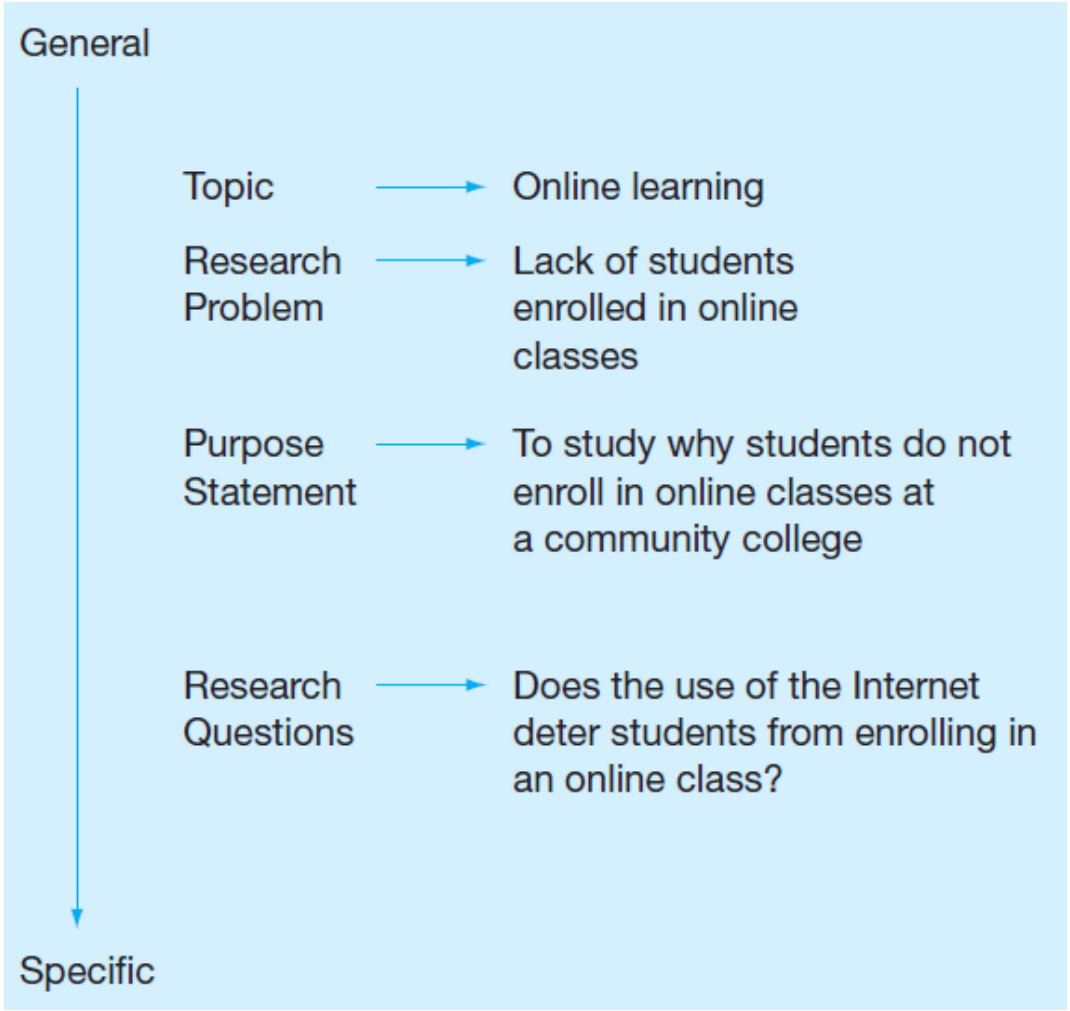
❖ Definition of Research Problem

➢ A clear and well defined problem statement should bring the following questions:

- ✓ What are the limitations/weaknesses in the existing research subject?
- ✓ What is the database required?
- ✓ What are the characteristics of data, which are relevant and need to be studied?
- ✓ What parameters need to be explored?
- ✓ What are the existing methods/techniques available for this purpose?

❖ Definition of Research Problem

➢ Example:



❖ Sources of Research Problems

➢ Following points need to be considered for defining the research problem

- ✓ Available literature need to be studied
- ✓ Nature of the problem need to be studied
- ✓An idea is generated after the discussion with colleagues, friends, seniors and experts in the research subject
- ✓ The research problem need to be rephrased into a working proposition

❖ Sources of Research Problems

- Most research in the humanities revolves around four Ps
  - ✓ To select a group of individuals (a group of individuals – or a community as such –'people'), to examine the existence of certain issues or problems relating to their lives
  - ✓ To ascertain their attitude towards an issue ('problem')
  - ✓ To establish the existence of a regularity ('phenomenon')
  - ✓ Or to evaluate the effectiveness of an intervention ('programme')

❖ Sources of Research Problems

➢ Aspects of a research problem

Aspects of a study	About	Study of	
Study population	People	Individuals, organisations, groups, communities	They provide you with the required information or you collect information from or about them
Subject area	Problem	Issues, situations, associations, needs, population composition, profiles, etc.	Information that you need to collect to find answers to your service research questions
	Programme	Contents, structure, outcomes, attributes, satisfaction, consumers, providers, etc.	
	Phenomenon	Cause and effect, relationships, the study of a phenomenon itself, etc.	

❖ Steps in Formulating a Research Problem

➢ The following points should be considered by the researcher in selecting the research problem:

- ✓ Too narrow or too vague statements for the research problem should be avoided
- ✓ Researchers should avoid the selection of controversial subjects for the research problem
- ✓ Research subjects that are done by a number of researchers generally avoided for the selection of research problem

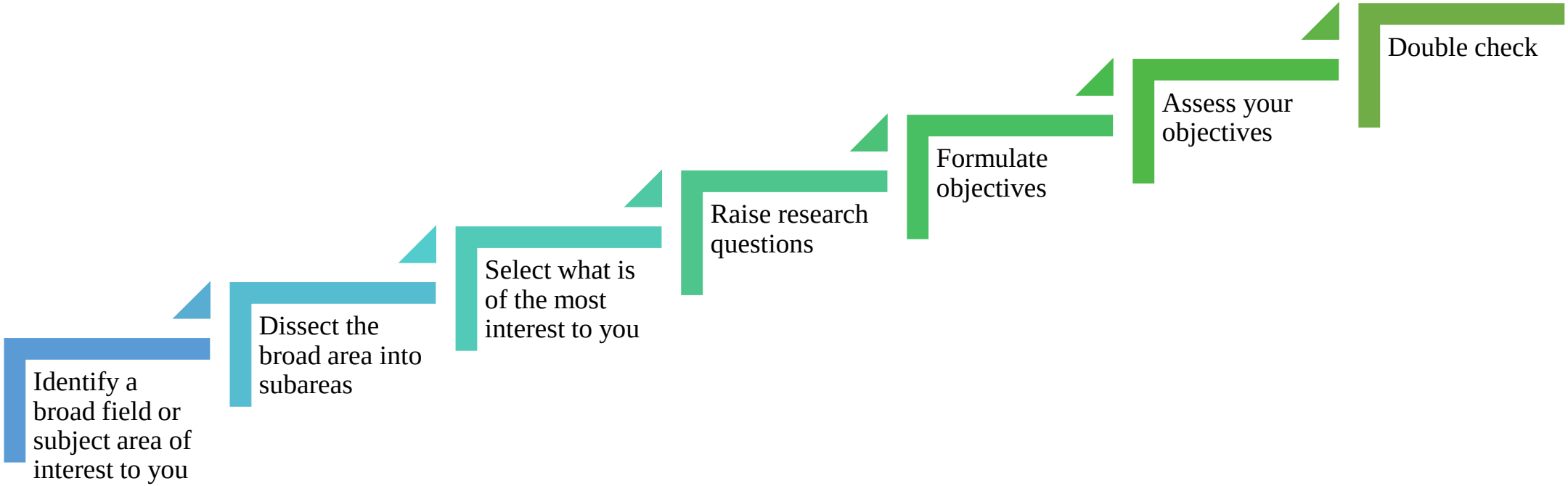
❖ Steps in Formulating a Research Problem

➢ The following points should be considered by the researcher in selecting the research problem (Cont.):

- ✓ The research problem should be feasible for experimentation/implementation
- ✓ Sufficient numbers of publications/materials should be available for finding the limitations/weaknesses of the subject
- ✓ The researcher's qualification, the importance of the subject, costs involved for experimentation and time factors should be considered

❖ Steps in Formulating a Research Problem

➢ The process of formulating a research problem consists of a number of steps:



❖ Steps in Formulating a Research Problem

➢ Example: Suppose you want to conduct a study in the area of health

Step 1	Step 2	Step 3	Step 4
Identify	*Dissect*	*Select*	*Raise questions*
Health	1 health services provided to the community 2 effectiveness of the services 3 cost of the services 4 health insurance schemes available to people 5 training of health professionals 6 adherence to ethics in health practices 7 attitude of the consumers towards health services 8 community responsiveness in the delivery of health services, etc.	Community responsiveness in the delivery of health services	1 How do the health administrators, planners, service providers and consumers define community responsiveness? 2 How can community responsiveness be achieved? 3 What indicators can be used to evaluate the effectiveness of community responsiveness strategies?

❖ Steps in Formulating a Research Problem

➢ Example: Suppose you want to conduct a study in the area of health

Step 5	Step 6	Step 7
Formulate objectives	*Make sure*	*Double-check*
Main objective: to evaluate the effectiveness of community responsiveness strategies in the delivery of health services Specific objectives: 1 to find out the understanding of the concept 'community responsiveness' among health administrators, planners, service providers and service consumers 2 to identify the strategies to implement the concept of community responsiveness in health services 3 to develop a set of indicators to evaluate the effectiveness of the strategies used in implementation of community responsiveness	Assess your objectives in the light of: 1 the work involved 2 the time available to you 3 the financial resources at your disposal 4 your and (your research supervisor's) technical expertise in the area	1 that you are really interested in the study 2 that you agree with the objectives 3 that you have adequate resources

❖ Steps in Formulating a Research Problem

➢ The researcher has to ask the following questions to himself before the selection of the final research problem:

- ✓ Whether sufficient resources are available for the researcher to carry out the research
- ✓ Budget provided/available for the research
- ✓ Cooperation from other colleagues/faculty to carry out the research work

❖ Errors in Defining the Problem Statement

➢ These errors are generally happened:

- ✓ Researchers don’t have an idea about what exactly to do (pinpointing of the research subject)
- ✓ Researchers are not qualified in the subject to carry out research design
- ✓ Why you feel that the topic is innovative or meaningful or of interest to the research
- ✓ Availability of database for the selected research subject
- ✓ Any legal issues related to the research domain

❖ Errors in Defining the Problem Statement

➢ These errors are generally happened (Cont.):

- ✓ Any issues related to the safety (such as military applications, etc.)
- ✓ No consideration of the cost involved while pursuing the research
- ✓ Too narrow or too vague problem statement
- ✓ Don’t select a controversial problem
- ✓ Don’t choose a subject/topic that is overdone; it is difficult to throw new light on such cases

❖ Quantitative and Qualitative Research Problem

- **Explaining or predicting** relations among ***variables*** is an important characteristic of quantitative research.
- **Exploring and understanding** a ***problem*** is a characteristic of qualitative research.



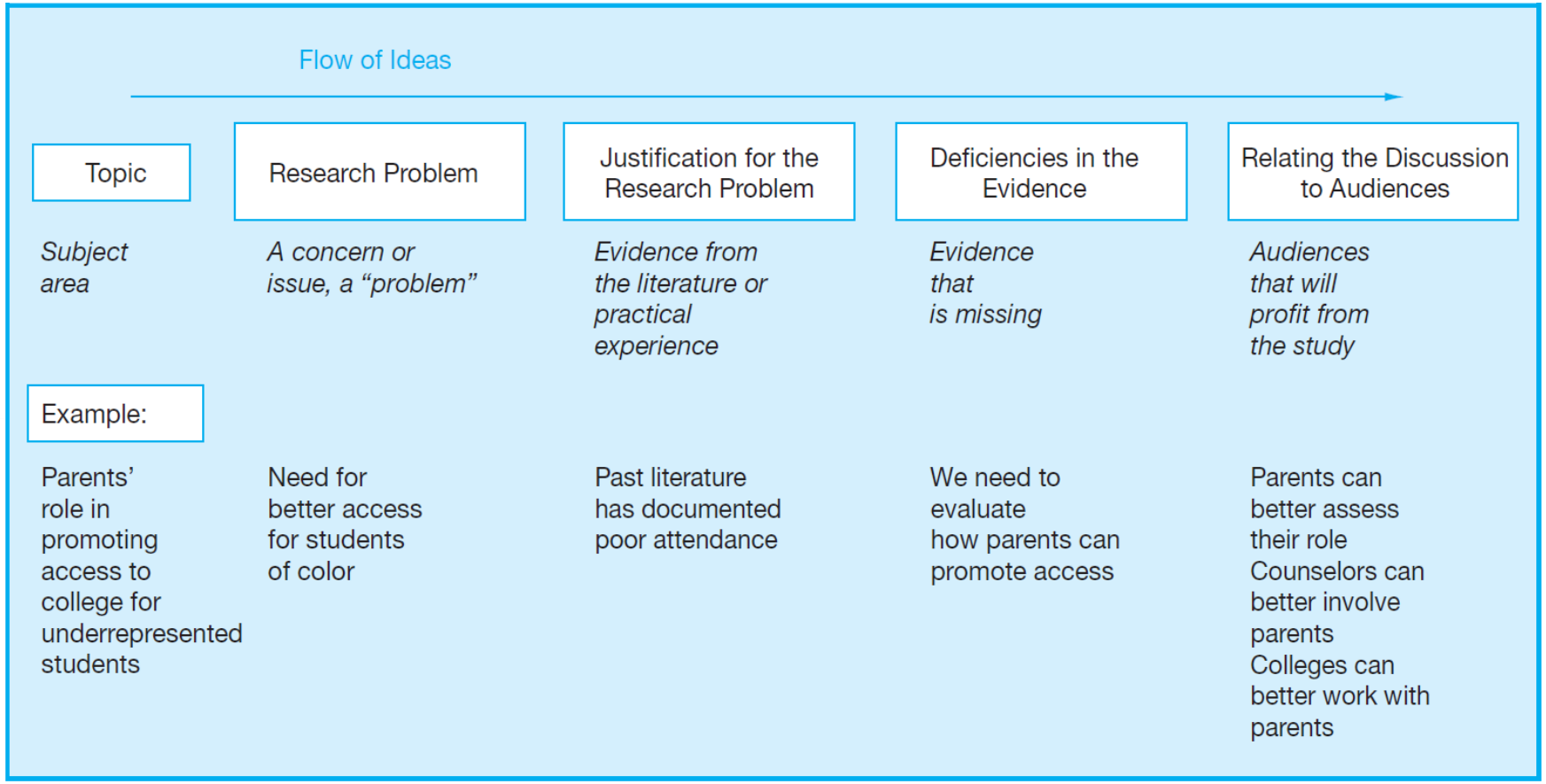
Image Source: Qualitative and Quantitative Research

❖ Quantitative and Qualitative Research Problem

Tend to use quantitative research	Tend to use qualitative research
❖ Measure variables ❖ Assess the impact of these variables on an outcome ❖ Test theories or broad explanations ❖ Apply results to a large number of people	❖ Learn about the views of individuals ❖ Assess a process over time ❖ Generate theories based on participant perspectives ❖ Obtain detailed information about a few people or research sites

❖ Elements of Writing a “Statement of the Problem”

➢ The statement of the problem has five aspects



❖ Elements of Writing a "Statement of the Problem"

➢ Sample "Statement of the Problem" Section

Statement of the Problem Elements	Exploring the Conceptions and Misconceptions of Teen Smoking in High Schools: A Multiple Case Analysis
The Topic	Tobacco use is a leading cause of cancer in American society (McGinnis & Foefe, 1993). Although smoking among adults has declined in recent years, it has actually increased for adolescents. The Center for Disease Control and Prevention reported that smoking among high school students had risen from 27.5 percent in 1991 to 34.8 percent in 1995 (USDHHS 1996). Unless this trend is dramatically reversed, an estimated 5 million of our nation's children will ultimately die a premature death (Center for Disease Control, 1996).
The Research Problem	Previous research on adolescent tobacco use has focused on four primary topics. Several studies have examined the question of the initiation of smoking by young people, noting that tobacco use initiation begins as early as junior high school (e.g., Heishman et al., 1997). Other studies have focused on the prevention of smoking and tobacco use in schools. This research has led to numerous school-based prevention programs and intervention (e.g., Sussman, Dent, Burton, Stacy, & Flay, 1995). Fewer studies have examined "quit attempts" or cessation of smoking behaviors among adolescents, a distinct contrast to the extensive investigations into adult cessation attempts (Heishman et al., 1997).
Evidence Justifying the Research Problem	

❖ Elements of Writing a “Statement of the Problem”

Deficiencies in Evidence	Of interest as well to researchers studying adolescent tobacco use has been the social context and social influence of smoking (Fearnow, Chassin, & Presson, 1998). For example, adolescent smoking may occur in work-related situations, at home where one or more parents or caretakers smoke, at teen social events or at areas designated as “safe” smoking places near high schools (McVea et al., in press). Minimal research attention has been directed toward the social context of high schools as a site for examining adolescent tobacco use. During high school students form peer groups which may contribute to adolescent smoking. Often peers become a strong social influence for behavior in general and belonging to an athletic team, a music group, or the “grunge” crowd can impact thinking about smoking (McVea et al., in press). Schools are also places where adolescents spend most of their day (Fibkins, 1993) and are available research subjects. Schools provide a setting for teachers and administrators to be role models for abstaining from tobacco use and enforcing policies about tobacco use (O’Hara et al., 1999). Existing studies of adolescent tobacco use are primarily quantitative with a focus on outcomes and transtheoretical models (Pallonen, 1998). Qualitative investigations, however, provide detailed views of students in their own words, complex analyses of multiple perspectives, and specific school contexts of different high schools that shape student experiences with tobacco (Creswell, in press). Moreover, qualitative inquiry offers the opportunity to involve high school students as co-researchers, a data collection procedure that can enhance the validity of student views uncontaminated by adult perspectives.

❖ Elements of Writing a “Statement of the Problem”

The Audience	By examining these multiple school contexts, using qualitative approaches and involving students as co-researchers, we can better understand the conceptions and misconceptions adolescents hold about tobacco use in high schools. With this understanding, researchers can better isolate variables and develop models about smoking behavior. Administrators and teachers can plan interventions to prevent or change attitudes toward smoking, and school officials can assist with smoking cessation or intervention programs.

Group Discussion

Prepared by: CHEN SOVANN

Q&A